Design of MgH_2-Ni-2D Nanosheets Composites for Enhancing Hydrogen storage/release property

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Abstract: This research aims to explore and improve the kinetics and hydrogen storage capacity of magnesium hydride MgH_2 by adding Ni catalyst through ball-milling a mix of MgH_2 and nickel complex salts to form Ni-MgH_2 nanocomposite. Currently, we are choosing 3 nickel complex salt candidates. Furthermore, to prevent agglomeration of MgH_2 nanocrystals and degradation effects created by ball milling and improve the cycle performance of Mg/MgH_2, a two-dimensional nano-sheet such as hydrogen boride(HB) or graphene will be added later to the composite. The performance of ball-milled MgH_2-Ni and MgH_2-Ni-HB/graphene composite will then be compared.

Keywords: Hydrogen storage, Solid-state hydrogen storage, MgH_2, Magnesium hydride, Nickel catalyst, Ni, Nickel-complex salt,ball milling, hydrogen boride/graphene, nanoconfinement

1 Introduction

Hydrogen is widely identified as a promising and clean energy carrier in the future, and as an attractive substitute to fossil fuel, to mitigate increasing environmental (achieving carbon neutrality goal) and sustainability concerns.[1,2,3] Solid-state hydrogen storage is a promising hydrogen storage method due to its higher gravimetric/volumetric density, safety and economy. [1]Hydrogen storage via absorption in metal hydrides offers high volumetric energy densities and increased safety due to hydrogen being chemically bound at lower pressures . [4] Among metal hydrides, MgH_2 is a promising hydrogen storage material that is widely covered by researchers due to their abundance, high hydrogen density (up to 7.6 wt %), thermodynamically stable, and can be safely transported . [4,5] Despite their advantages, challenges include slow hydrogen release, decreased absorption with cycles, and the need for high temperatures for release (>300 °C). [1,4] Utilizing two-dimensional materials like graphene as a substrate and loading a catalyst with them has been explored to address these issues. [1]

Catalysts such as transition metal catalysts (such as nickel Ni) has been considered as one of the most effective stategies to improve the hydrogen storage performance of Mg/MgH_2. As a metal catalyst, Ni has been reported to improve repeatability and decrease hydrogen release and absorption temperatures. [1] It can effectively accelerate the dissociation and combination of hydrogen atoms and decrease the activation energy for hydrogen absorption, thus improving the dehydrogenation kinetics and lowering the dehydrogenation temperatures. Therefore, the doped catalyst should be effectively improved to promote the dehydrogenation kinetics of MgH_2 as much as possible, to maintain the high storage capacity of Mg/MgH_2. The reduction in size and improvement of disperity would increase the catalytic activity of the catalyst [4]. The addition of a 2D-nanomaterial, such as carbon-based materials (graphene, carbon nanotubes) or hydrogen boride (HB) can be used to inhibit the grain aggregation and growth of Mg/MgH_2 during cycle-life (kinetics). [1,4] Supporting MgH_2 in a clustered state on two dimensional materials is expected to prevent degradation of hydrogen release and absorption capacity due to aggregation of Mg and catalyst. [1]

Thus, a suitable transition catalyst precursor and a 2D-nanomaterial could be introduced to form the Mg/MgH_2 matrix to form a catalyst with high catalytic activity, thereby improving the dehydrogenation kinetics and cycle stability (e.g over 7 wt %) [4]. The catalyst precursor for this research will be a nickel complex salt, and the 2D nanomaterial will be HB or graphene.

MgH_2, a hydride of Mg abundant in the earth, shows a high hydrogen storage capacity, but a lower hydrogen release temperature is required. Here, this research aim to decrease the temperature of MgH_2 for hydrogen release, improve the hydrogenation kinetics of MgH_2, and to improve hydrogen absorption/release cycle stability, and enhance storage capacity of MgH_2.

In this research ball-milling, a technique used to decrease the particle size in metal hydrides [6], will be used to 1) dope the Mg/MgH_2 with the Ni through mixing MgH_2 with nickel complex salt and 2) add HB/graphene to the MgH_2-Ni composite through mixing with the MgH_2 and nickle complex salt. Studies demonstrate that ball milling enhances the hydrogen storage properties of metal hydrides.[3] Yuan et al., with different ball-milling time duration, demonstrated that ball-milling of TiH_2, significantly lowered the first-stage H_2 release temperature by approximately 150 °C and improved hydrogen release kinetics. [3,7]Ball milling is a convenient technique to introduce additives in complex metal hydrides. For instance, ball milling was used to introduce metal and metal halides additives to $LiAlH_4$ and $LiBH_4$, significantly reducing the activation energy for hydrogen release. [6]

We hypothesise that ball-milling MgH_2 and nickel complex salt to form MgH_2-Ni composite with nickel as a catalyst will improve hydrogen storage capacity of MgH_2, decrease the temperature for hydrogen release of MgH_2, and improve the kinetics compared to pure ball-milled MgH_2. Smaller MgH_2 particles improve the hydrogen release of MgH_2. We may also add a 2D-dimensional material such as HB /graphene to prevent agglomeration of MgH_2 and Ni particles and improve cycle stability.

2 Related Works

Shen et. al[4] developed a facile one-step high energy ball milling to in situ form ultrafine Ni nanoparticles catalyst in the MgH_2 matrix, combining the nickel acetylacetonate complex as a precursor and extended grahite (EG). The in situ formed ultrafine Ni particles catalyst from the Ni(acac$_2$) can significantly improve the desorption kinetics of MgH_2. On the other hand, the cycle performance of Mg/MgH_2 is improved by EG. They reported the formed MgH_2-Ni-EG nanocomposite with the optimized doping amounts of Ni and EG can release an exceptional 7.03 wt% H_2 within 8.5 min at 300 °C after 10 cycles. The exceptional hydrogen storage performance was credited to a 26.9% decrease in the dehydrogenation activation energy in comparison with the pure MgH_2· In addition, at 50 °C, it can also absorb 2.42 wt% H_2 within 1 hour. Thus, their work provides a methodology to significantly improve MgH_2hydrogen storage performance.

Galey et. al[2] added Ni_3PCy_3 complex into MgH_2. Different composites were prepared by planetary ball-milling by doping MgH_2 with (i)$PCy_3$with/out nickel nanoparticles, (ii) different $NiPCy_3$ contents (5-20 wt %) and (iii) nickel and iron nanoparticles with/out $NiPCy_3$. All MgH_2 composites showed much better dehydrogenation properties than the pure ball-milled MgH_2· The hydrogen absorption/release kinetics of the Mg/MgH_2 system were significantly enhanced by doping with only 5 wt% of $NiPCy_3$(0.42 wt% Ni); the mixture desorbed H2 starting at 220 °C and absorbed 6.2 wt% of H_2 in 5 mins. at 200 °C under 30 bars of hydrogen. This remarkable storage performance was not preserved upon cycling due to the complex decomposition during the dehydrogenation process. The hydrogen storage properties of $NiPCy_3-MgH_2$ were improved and stabilized by the addition of Niand Fe nanoparticles. The formed system released hydrogen at temperatures below 200 °C, absorbed 4 wt% of H_2 in less than 5 min at 100 °C, and presented good reversible hydriding/dehydriding cycles. A study of the different storage systems leads to the conclusion that the $NiPCy_3$ complex acts by restricting the growth of the crystal size of Mg/MgH_2,catalyzing H_2 release and homogeneously dispersing nickel on the Mg/MgH_2 surface.

These studies suggest that combining MgH_2 with a transition metal-catalyst precursor such as Ni complex salts can result in a Ni-MgH_2 composite with better hydrogen absorption/release and storage kinetics, and the addition of a 2D-nanomaterial/nanosheet can improve cycle stability.

In her Master's thesis, Noguchi created MgH_2-Ni/HB composites and investigate their hydrogen storage-release mechanism[1]. Ni nanoclusters were successfully dispersed on HB sheets. MgH_2-Ni/HB composite shows lower H_2 release temperature than ball-milled MgH_2. Specifically, the H_2release temperature of $MgH_2-(Ni_{0.01}/(HB)_1)$ @Ni2.0wt% was 550 K, about 50 K lower than that of ball-milled commercial MgH2 (600 K). The use of $MgH_2-(Ni_{0.05}/(HB)_1)$ and $MgH_2(Ni_{0.005}/(HB)_1)$ were also tried, but $MgH_2-(Ni_0 \cdot {}_1/(HB)_1)$ resulted in the lowest hydrogen release temperature at the lowest Ni content (2wt%). They hypothesized that MgH_2 supported on Ni/HB would prevent MgH_2 aggregation

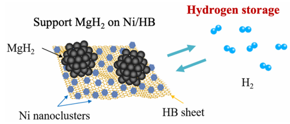

Figure 1: *Schematic image of the conception of Noguchi's work in her master's thesis [1]*

and lower the H_2 release temperature due to the catalytic effect of Ni. They proposed a model in which Ni nanoclusters act as a catalyst for hydrogen recombination and the MgH_2-Ni interface plays an important role in lowering the hydrogen release temperature. MgH_2-Ni/HB composites with $Ni_{0.005}$/HB, $Ni_{0.05}$/HB, and $Ni_{0.1}$/HB were synthesized and their hydrogen release and absorption properties were investigated. The results showed that the hydrogen release temperature was lowered by 50 K at the lowest Ni 2.0wt% when $Ni_{0.1}$/HB was used. The hydrogen desorption profiles also indicated that the hydrogen desorption temperature decreases in a stepwise sequence. This suggests that the contact between Ni which acts as a hydrogen recombination catalyst and MgH_2, is good for H-hopping in a stepwise sequence depending on the Ni content. The hydrogen absorption properties of MgH_2-$Ni_{0.05}$/HB @Ni1wt% were investigated using DSC measurements. Hydrogen desorption occurs when the temperature is raised to 350 ℃, and hydrogen absorption occurs when the temperature is lowered to 50 ℃.

However, these researches focus on only one type of Ni complex salt. We propose mixing MgH_2 with different complex salts precursors and add/not add HB or graphene. Then we compare the hydrogen storage and kinetics of the different MgH_2-Ni composites under the same conditions (same ball-milling parameters and same HB/graphene). The performance of Mg-Ni composites and the Mg-Ni-HB/graphene will be compared. As of now, we have 3 Ni-complex salt candidates. The parameters we use to choose different Nickel complex salt candidates are 1) the nearest atom/ion/compound to the Ni in the nickel complex candidates 2) size (molecular weight) and 3) commercial (such as sold by Sigma Aldrich or Fujifilm) for practical purposes. The type of element/compound that is nearest to Ni is considered for specific catalyst binding characteristics and dehydrogenation/ hydrogenation characteristics.

3 Materials for Investigation

3.1 Candidate Nickel Complex Salts

Possible Nickel Complex salts discussed and suggested are such as Ni(acac)$_2$, $NiCl_2$, or $Ni(NO_3)_2$. Using the parameters mentioned above to choose the candidates, as of now we have 3 commercial candidates available in TCI Chemicals (Bis(1,5-cyclooctadiene)nickel(0)) and Fujifilm (Nickel(II) Hydroxyacetate and Nickel (II) phtalocyanine). Currently we are on the process of inquiring the availability and obtaining the nickel complex salt candidate samples.

I. Nickel(II) phthalocyanine(($C_{32}H_{16}N_8$)Ni), min. 94% The available sample we consider has a mass of 5 g. The molecular structure of the complex can be seen in Figure 2 above.

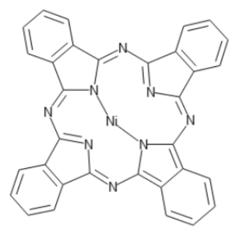

Figure 2: *Structure of Nickel(II) phthalocyanine*

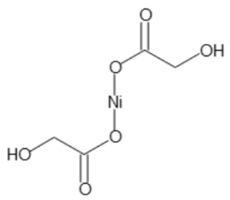

Figure 3: *Structure of Nickel(II) Hydroxyacetate*

The $(C_{32}H_{16}N_8)Ni$ sample candidate complex has characteristics:
1) The element closest to Ni is N(Nitrogen)
2) The molecular weight is 571.23 amu

II. Nickel(II) Hydroxyacetate ($Ni(OOCCH_2OH)_2$)
The available sample we consider has a mass of 5 g. The molecular structure of the complex can be seen in Figure 3 above. The $Ni(OOCCH_2OH)_2$ sample candidate has characteristics:
1) The element closest to Ni is O(Oxygen)
2) The molecular weight is 208.75 amu

III. Bis(1,5-cyclooctadiene)nickel(0) ($C_{16}H_{24}Ni$), 5 grams
The available sample we consider has a mass of 5 g. The molecular structure of the $C_{16}H_{24}Ni$ complex can be seen in Figure 4.

The sample candidate complex has characteristics:
1) The element closest to Ni is C (Carbon)
2) The molecular weight is 275.06 amu

Before we acquire these commercial nickel complex salts, we may first ball mill and do characterisation on Nickel(II) Chloride and Nickel (II) acetylacetonate ($Ni(acac)_2$) first.

Nickel (II) acetylacetonate ($Ni(C_5H_7O_2)_2$) has the following characteristics:

1. The compound closest to Ni is acetylacetone, composed of CH_3andO
2. The molecular weight is 256.91 amu

The molecular structure of $Ni(acac)_2$ can be seen in Figure 5.

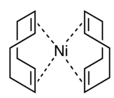

Figure 4: *Structure of Bis(1,5-cyclooctadiene)nickel(0)*

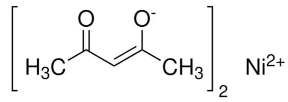

Figure 5: *Structure of Ni(acac)2*

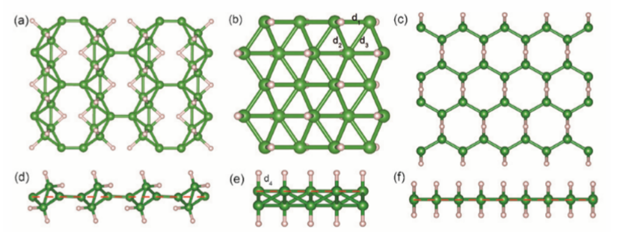

Figure 6: *Structure of stabilized hydrogen boride sheets (Green: B, Pink: H) (a) (c) Image viewed from above the seat surface, (d) - (f) Side view of the sheet (a), (d) C2/m, (b), (e) Pbcm, (c), (f) Cmmm. Image is taken from [1]*

3.2 Hydrogen Boride(HB) nanosheets

Boron is an element that has attracted much attention because of its low density, high mechanical strength, high heat resistance, and high melting point. [1] MgB_2 is a binary compound composed of hexagonal boron sheets alternating with Mg cations [8], as in Figure 7. Because MgB_2 inherently contains 2D boron sheets, Kondo et. al considered that the designed ion-exchange method between Mg cations and other cations is a way to produce well-defined, borophene-related, and stable 2D materials through a top-down approach. In 2017, Kondo et. al demonstrated the experimental formation of HB sheets by using cation-exchange between protons and Mg cations of MgB_2 in nonaqueous media at room temperature. [8]

As reported by Kondo et.al [8] in 2017, the use of ion-exchange resins for MgB_2 enabled them to synthesize hydrogen boride sheets, which are two-dimensional nanosheets composed of hydrogen and boron. The synthesis process is showed in Figure 7.By reacting MgB_2 with a strongly acidic cation exchange resin in acetonitrile or methanol, magnesium ions, and protons are ion-exchanged, and then exfoliated, two-dimensional nanosheets (hydrogen boride sheets) were successfully synthesized. This entire process was conducted under nitrogen at room temperature and ambient pressure with stirring.After 3 days, black precipitates were removed by filtration, and the filtrate was dried on a hot plate at 343 K under nitrogen. HB sheets were then obtained as a yellow powder as the ion exchange reaction product at an average reproducible yield of 42.3%. The main byproduct was $B(OH)_3$, whichwas formed by the hydrolysis reaction of HB sheets with waterabsorbed in the ion-exchange resin . The HB sheets were purified by filtration after cooling the acetonitrile suspension to 255 K . In the methanol suspension, $B(OH)_3$ reacts with methanol to form trimethyl borate, which evaporates under the drying process at 343 K.

The boron is negatively charged, and the hydrogen is positively charged, indicating that the hydrogen in the nanosheet is not hydride but exists with a proton-like positive charge.

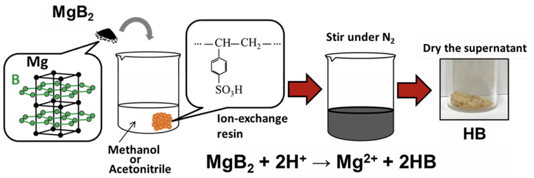

Figure 7: *Synthesis process of hydrogen boride sheets. Image is from [8]*

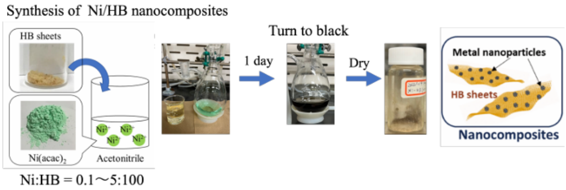

Figure 8: *Schematic image of synthesis method of Ni/HB sheets. Image is from[1]*

Therefore, our laboratory named the synthesized nanosheet as "Hydrogen Boride." The structure of this borohydride is very similar to that of stable borohydride proposed in a previous study (Fig 6), and the thickness of the borohydride is less than 0.5 nm. [1] Hydrogen is bonded to the honeycomb structure of boron in a bridging structure, and this structure is found to be localized rather than long-periodic. Boron atoms form a hexagonal 2D network in HB sheets, wherein hydrogen atoms are bound to boron atoms by three-center two-electron (B–H–B) and two-center two-electron (B–H) bonds, with chemical stability against water [9]. The ratio of hydrogen and boron is approximately 1:1. [8]

HB sheets have been experimentally verified to exhibit excellent solid acid catalytic activity and highly sensitive gas-sensing capability, as well as semimetal electronic, light-responsive hydrogen release, and carbon dioxide adsorption and conversion properties [9].

Previous experiments in our laboratory have revealed that hydrogen boride sheets can reduce metal ions in liquids and support metal nanoparticles. Specifically, six kinds of metal ions, Pd, Pt, Ni, Co, Zn, and Cu, have been investigated, and it has been shown that metal ions with redox potentials lower than Ni can be reduced but those with redox potentials higher than Co cannot. [1] A group in China reported that highly dispersed carbon-supported Pt nanoparticle electrocatalysts can be synthesized by using HB sheets without using surfactants or other reducing agents. The catalytic activity and stability of the product are superior to those of commercial catalysts for fuel cell cathodes. [1,10] In this research, either HB or graphene will be chosen as the 2D-material to support the Ni catalyst and MgH_2.

As a future reference, Noguchi [1] synthesised Ni/HB using the following process with Powder HB 95 mg / acetonitrile 80 mL. (Figure 8)
$Ni(CH_3COCHCOCH_3)_2$ 206.6 mg, 102.7 mg, 10.7 mg / acetonitrile 80 mL
1. HB acetonitrile dispersion and Ni complex acetonitrile solution were prepared by adding each to acetonitrile.
2. The synthesis was started by adding HB acetonitrile dispersion to Ni complex acetonitrile solution.
3. In order to degas the air in acetonitrile, "pull vacuum for one minute → introduce argon gas to normal pressure" is repeated for three sets.
4. Stirred at 310 rpm for about 24 hours under an Ar atmosphere. The mixture was covered with a double layer of black plastic bags to shield it from light.
5. After about 24 hours, the stirrer was removed and concentrated and dried while pulling a vacuum. During the concentration, the mixture was immersed in an oil bath at 30-45°C to avoid frost formation. During drying, the temperature was increased by 5 ℃ in increments to 70 ℃.

3.3 Graphene and Nanoencapsulation

Schneemann et.al reported in their review that materials that have been used for the nano-encapsulation of particles are graphene, graphene oxide, and poly(methyl methacrylate) polymers. Essentially, hydrogen has small enough kinetic diameter to diffuse through these wrappers, while other gases are blocked out. These wrappers protect the materials from external contamination, stabilize small particles, and tune the hydrogenation and desorption behavior. [6]

4 Methodology

The proposed research project is designed to comprehensively investigate hydrogen storage capacity, hydrogen absorption-desorption behaviour, and kinetics of MgH_2-Ni composite from nickel complex salts formed through ball-milling. This study also aims to explore the fundamental mechanisms by which ball milling, a process of mechanical alloying, influences the microstructural, morphological, and hydrogen storage characteristics of the MgH_2-Ni composite. The catalyst activity of Ni and the role of MgH_2-Ni interface will also be investigated.

A 2D-nanosheet (HB or graphene) will be added later to form MgH_2-Ni-HB/graphene composites with each different nickel complex salt . Then the hydrogen storage capacity, hydrogen absorption-desorption behaviour, kinetics, Ni catalyst activity, role of MgH_2-Ni, effect of addition of HB/graphene, and reaction pathways of the MgH_2-Ni-HB/graphene composite will be compared with MgH_2-Ni-HB/graphene.

4.1 Ball-milling

Ball-milling has been extensively explored for nanosizing and nanostructuring of metal-hydrides, due to its relative simplicity and general applicability of the process. It is also a versatile technique to introduce dopants and catalysts. Ball-milling, a mechano-chemical process, involves the mechanical grinding of a solid sample with one or more inert and hard balls (e.g., ceramic, flint, tungsten carbide, or stainless steel) moving at a high speed and crushing the powder. In practice, as particles decrease in size upon ball-milling, a certain limit is reached beyond which no reduction in size occurs. Typically, the grain size of metal hydrides decreases from a few microns to a few nanometers. Ball milling is efficient in reducing the activation energy of hydrogen absorption/desorption from MgH_2. However, this improvement is not always due to the small particle size. Ball-milling complex metal hydrides is less efficient in reducing the particle size, compared to doing so for binary and intermetallic hydrides. In addition, ball milling is a convenient technique to introduce additives in complex metal hydrides.

However ball milling has important limitations, such as yielding nonuniform particle sizes and shapes as well as issues with contamination. In addition, metal and metal hydride ball milling is believed to create highly reactive surfaces that readily react with oxygen, even residual oxygen present on the surfaces of the milling balls and vessel. Although the exact role of such oxide layers is not very well-understood, excessive oxidation can cause capacity loss and create issues with hydrogen diffusion. On the other hand, the resulting high density of grain boundaries can, to some extent, facilitate hydrogen diffusion and thus improve the overall kinetics. Finally, even though the hydride particles are in the desired nanoscale regime initially, the material can undergo significant recrystallization and particle growth upon heating during hydrogen absorption and desorption, eventually leading to bulk-like behavior.

Many recent experiments suggest that nanostructured metal hydrides have significantly, rather than incrementally, different thermodynamics and kinetics than their bulk counterparts. For example, a substantial decrease in the H_2 desorption enthalpy is predicted for very small MgH_2 particles. Similarly, force field modeling predicts a decrease in thermodynamic stability of MgH_2 as particle size decreases from 2.0 to 0.6 nm. Nanoconfinement of Li_3N was shown to fundamentally alter both the hydrogenation and dehydrogenation reaction pathways as a direct consequence of solidsolid nanointerfaces within the material. [6] Furthermore, the reduction in size and improvement and improvement of dispensity would increase the catalytic activity of the catalyst. [1] Thus, for research purposes, nanostructured MgH_2 particles and Ni particles might be desired rather than their bulk counterparts. [6]

4.2 Sample Preparation

Nickel complex salts (possibly the 3 candidates mentioned in Section 3.1, may be changed or determined later in future discussion), MgH_2, and HB or graphene (2D-material to be decided later with discussion) will be used as raw materials. First, MgH_2 and each of the Nickel complex salts will be mixed and ball-milled(a mechano-chemical process), to form MgH_2-Ni composite. We will possibly use a planetary ball mill of Retsch company. The variables of interest include milling time (to assess the impact of prolonged mechanical energy input), ball-to-powder ratio (to determine the optimal mass ratio for efficient milling) [3], and milling speed. The ball-milling time is 4 hours followed by 1 hour of washing,
ball milling speed is 400rpm
and the ball-to-powder ratio will be the same as the ratio used by Noguchi's previous research.

We will use two methods to synthesise MgH_2 and Ni in our experiment, and then compare the results using characterisation techniques. The two methods are the following:

1. Ball mill MgH_2 $\longrightarrow$ add Ni $\longrightarrow$ handmill

Table 1: *Characterisation techniques*

Characterisation techniques	
Method	Function/Overview
XRD	Crystallinity
IR	Bonding
SEM	Morphology
TG-DTA&MASS	H_2 release temperature,activation energy
PCT	Advanced H_2 capacity

2. Ball mill MgH_2 + Ni

Later, HB or graphene may also be introduced into MgH_2 with each of the nickel complex salt together and then ball-milled, to form $MgH_2-Ni-HB$/graphene composites.

4.3 Characterisation

To ensure a comprehensive analysis of the milled samples, an array of analytical techniques will be employed for this research (Table 1).

The purpose of each method is as of the following:

1. X-Ray Diffraction Method (XRD)
   - XRD is is a fundamental technique for characterizing the crystallographic structure, phase composition, and structural changes in hydrogen storage material XRD is widely used in hydrogen storage research to investigate the structural properties of metal hydrides, complex hydrides, and other crystalline storage materials. It allows for the identification of the hydrogen storage phases, the determination of the phase abundances, and the study of phase transitions during the hydrogen sorption processes. One of the advantages of XRD is its non-destructive nature, allowing for the characterization of the bulk properties of the material. [11]
2. Infrared Spectroscopy (IR)
   - Infrared spectroscopy is a method for estimating the identity and chemical structure of a substance by utilizing the fact that the infrared spectrum is unique to the substance. Infrared absorption spectra can be obtained by irradiating a substance with infrared light and measuring the intensity of the transmitted or reflected infrared light. When infrared light is irradiated to a molecule, it resonates with the molecule's natural vibration and absorbs it. Infrared light which are absorbed based on the electronic transitions of materials, is also called the vibrational spectrum, because it is absorbed based on the vibrational and rotational motions of molecules, which have less energy than the electronic transitions. The electron energy has the largest energy level spacing, followed by the vibrational energy, and the rotational energy has the smallest energy level spacing. The absorption spectrum of a molecule is observed because the

energy of light absorbed by the molecule causes transitions from lower energy states to higher energy states beyond these intervals.[1]

- Absorption does not necessarily occur when the energy of infrared light corresponds to the energy level difference. There are two selection rules, one of which is that infrared absorption occurs only when the dipole moment changes due to molecular vibration. For vibrational transitions to occur, it is necessary for transition dipole moments to be generated by molecular vibrations. From the above, infrared absorption does not occur for monatomic molecules such as oxygen and nitrogen. In addition, if the dipole moment is symmetrical at the surface of the electrode, the dipole cancels out and no absorption takes place.

3. Scanning Electron Microscopy
    - Scanning electron microscopy (SEM) is an ideal tool for studying the microstructure and surface characteristics of hydrogen storage materials due to its high-resolution imaging capabilities. SEM can reveal detailed morphological features of materials, helping to understand the interactions between hydrogen and these materials, which is crucial for designing more efficient hydrogen storage systems. The microstructure of materials, such as pore size, distribution, and surface roughness, directly affects the adsorption and diffusion rates of hydrogen. Through SEM, researchers can clearly see these structural features and evaluate their specific impact on hydrogen storage performance. For example, larger pores may promote rapid hydrogen diffusion, while higher surface roughness can increase the surface area, providing more active sites for hydrogen adsorption. Additionally, SEM analysis can reveal potential defects on material surfaces, such as cracks, fractures, or other irregular shapes, which could affect the long-term stability and hydrogen storage efficiency of the materials. By regularly using SEM to monitor these materials, scientists can track performance changes during long-term use and adjust preparation processes or select more suitable materials accordingly. [11]

4. Thermogravimetric- Differential Thermal Analysis (TG-DTA) and Mass Spectroscopy (MASS)
   As TG-DTA and MASS are connected, we can do these characterisation techniques at the same time.
    - a) TG-DTA
        - TG-DTA is a simultaneous thermogravimetry (TG: Thermogravimetry) and differential thermal analysis (DTA: Differential Thermal Analysis) system. It measures the weight change and thermal change of a sample when its temperature is changed. Thermogravimetry is performed using a thermobalance. As the temperature rises, convection and buoyancy forces around the balance cause an apparent weight change. In the differential type, buoyancy and convection forces are applied to both the sample holder and the reference holder, and these effects are canceled out.Differential thermal analysis (DTA) detects the thermal change in a sample due to temperature change as a temperature difference from a reference material, and the DTA signal provides information on the transition temperature and reaction temperature of the sample, and whether the transition or reaction is endothermic or exothermic. [1]
    - b) MASS
        - MASS is a highly effective qualitative and quantitative analytical technique used to identify and quantify a wide range of clinically relevant analytes. Most mass spectrometry data are presented in units of the mass-to-charge ratio (m/z), where m is the molecular weight of the ion (in daltons), and z is the number of charges present on the measured molecule. For small molecules (<1000 Da), there is typically only a single charge; therefore, the m/z value is the same as the mass of the molecular io.n However, when larger molecules such as proteins or peptides are measured, they typically carry multiple ionic charges, and, therefore, the z-value is an integer greater than 1. The m/z value is a fraction of the ion's mass in these cases. [12]
        - Mass spectrometers convert molecules into ions, which are then manipulated using electric and magnetic fields. Atoms and molecules must first be ionized before being accelerated through the mass spectrometer and detected. [12]
        - A chart is generated to analyze the mass spectrometer's results, with the mass-to-charge ratio (m/z) on the x-axis and the relative intensity on the y-axis. For a given sample, the most abundant ion in the sample molecule is known as the base peak. This ion is set to 100% on the y-axis for its relative intensity, and all the remaining ion peaks are generated relative to this value. The molecular ion peak is known as the parent peak because it corresponds to the molecular weight of the sample. [12]

5. Pressure-Composition-Temperature curve (PCT)
    - The PCT curve is primarily used to describe the relationship between the composition of a chemical system and pressure under isothermal conditions, and it is commonly applied in the study of gas adsorption and metal hydrides for hydrogen storage. By analyzing the PCT curve, insights can be gained into the phase behavior of the system, adsorption or desorption processes, and the hy-

drogen storage capacity of the material. [11]

- The Vant Hoff equation

$$lnK = -\Delta H/RT + \Delta S/R \quad (1)$$

where K is the equilibrium constant, H is the standard reaction enthalpy (kJ/mol), S is the standard reaction entropy (J/mol·K), R is the gas constant (J/mol·K), and T is the temperature (K), reveals the relationship between the chemical equilibrium constant and temperature, forming the theoretical foundation for studying the effect of temperature on equilibrium in reaction thermodynamics.

- By experimentally determining the changes in the equilibrium constant with temperature, thermodynamic parameters such as enthalpy and entropy changes can be obtained. The combination of the PCT curve and the van't Hoff equation allows for in-depth study of thermodynamic behavior in multi-phase systems such as gas–solid and liquid–solid phases, providing reliable thermodynamic parameters to optimize material performance and understand reaction mechanisms. [11]
- As this technique is more advanced compared to the previous methods, the PCT may be an optional step in this research.

We will first do XRD and IR before and after ball milling. After ball milling, we will also do TG-DTA and TPD.

5 Conclusions and Future Research Plan

In this research proposal, we seek to investigate and enhance the hydrogen storage of MgH_2, through adding a Ni catalyst by ball-milling with nickel complex salt to form a MgH_2-Ni composite. We hypothesise that this will improve hydrogen storage capacity of MgH_2, decrease the temperature for hydrogen release of MgH2, and improve the kinetics. We may also add a 2D-dimensional material such as HB /graphene to prevent agglomeration of MgH_2 and Ni particles and improve cycle stability.

Hopefully, this will improve the feasibility of hydrogen storage by an abundant material to ensure a sustainable, renewable energy future.

Currently, we are on the process of inquring the availability and obtaining the chosen nickel complex samples. Obtaining samples may take a few days to a few weeks. After we acquire the nickel complex salts, we will first start with a careful evaluation of each candidate complex, including a thorough review of its chemical properties and safety data sheets (SDS) to establish proper handling protocols. Then, we will proceed with the nickel complexes sequentially. First, we ball mill MgH_2 and each of the nickel complexes salt sequentially for approximately 4 hours (a half-day process). The first ball-milled nickel complex will be $Ni(acac)_2$ to optimise the ball-milling conditions for preparing MgH–Ni composites. For $Ni(acac)_2$, we will perform the full experimental workflow: pre-characterization of the starting materials, ball milling, post-milling characterization, hydrogen desorption measurements, and post-desorption characterization. Each characterisation technique may take a day (such as XRD) to a few days or weeks. We will then analyse the results. Based on these results, robust and reproducible ball-milling conditions will be established. Once the standard ball-milling conditions have been determined, the same workflow will be applied sequentially to the other nickel complexes to enable consistent and reliable comparisons. In this comparative stage, we will primarily evaluate hydrogen desorption behavior using TG–DTA and TPD as a screening approach. Based on these results, only the most promising material(s) will be selected for more comprehensive hydrogen storage evaluation, including PCT measurements to assess hydrogen absorption/desorption performance.

After completing the comparative study of the MgH_2–Ni systems, graphene (or HB) will be introduced in a later stage of the project. The additive will be incorporated only into the most promising MgH_2–Ni composite(s), and the same experimental workflow will be repeated to evaluate its influence on hydrogen storage performance.

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